

Automate Checkmk with Ansible



René Koch
DI (FH) René Koch



About me

- René Koch
- Self-employed consultant for:
 - Red Hat Ansible (Automation Platform)
 - Red Hat Enterprise Linux
 - Red Hat Satellite
 - Red Hat Identity Management (IPA)
- Experienced monitoring user (Nagios, Icinga, Checkmk)



About me

- René Koch
 - rkoch@rk-it.at
 - +43 660 / 464 0 464
 - <https://www.linkedin.com/in/rk-it-at>
 - <https://github.com/rk-it-at>
 - <https://github.com/scrat14>



 **Please introduce yourself**



Timetable

- 09:00 – 10:30: Workshop
- 10:30 – 10:45: Break
- 10:45 – 12:30: Workshop
- 12:30 – 13:45: Lunch break
- 13:45 – 15:30: Workshop
- 15:30 – 15:45: Break
- 15:45 – 17:15: Workshop

Table of contents

-  Prepare LAB environment
-  Installation of Checkmk server
-  Installation of Checkmk agents
-  Checkmk server configuration
-  Fetch information with lookup plugins
-  Using Checkmk as Ansible inventory
-  Demo: Self-healing with Checkmk, Ansible Automation Platform and rulebooks



Preparing the LAB environment



Preparing the LAB environment

-  Create LAB environment
-  Ansible control node =
Checkmk server =
Checkmk agent
-  Use local connection
-  Configure sudo for privileged access

Preparing the LAB environment

-  Slides and code are available on GitHub:
https://github.com/rk-it-at/checkmk_conference_2026
- **+** Optional: install and clone the repository:

```
git clone https://github.com/rk-it-at/checkmk_conference_2026.git
```



Preparing the LAB environment

-  Available VMs (if you do not have your own)
-  Passwords: <TODO>



Preparing the LAB environment

Hostname	IP Address	Username	Participant
ansible-ws-1		ansible	
ansible-ws-2		ansible	
ansible-ws-3		ansible	
ansible-ws-4		ansible	
ansible-ws-5		ansible	
ansible-ws-6		ansible	
ansible-ws-7		ansible	



Preparing the LAB environment

Hostname	IP Address	Username	Participant
ansible-ws-8		ansible	
ansible-ws-9		ansible	
ansible-ws-10		ansible	
ansible-ws-11		ansible	
ansible-ws-12		ansible	
ansible-ws-13		ansible	
ansible-ws-14		ansible	
ansible-ws-15		ansible	

Preparing the LAB environment

-  Configure sudo for privileged access

```
$ su -  
# <EDITOR> /etc/sudoers.d/ansible
```

```
<USERNAME> ALL=(ALL) NOPASSWD: ALL
```

NOTE

Use the root user to run this command!

Replace <EDITOR> with vi, nano or your favorite editor

Replace <USERNAME> with your system user



LAB 1: Configure sudo

LAB 1: Configure sudo

-  Configure sudo for privileged access

```
$ su -  
# <EDITOR> /etc/sudoers.d/ansible
```

```
<USERNAME> ALL=(ALL) NOPASSWD: ALL
```

NOTE

Use the root user to run this command!

Replace <EDITOR> with vi, nano or your favorite editor

Replace <USERNAME> with your system user



Install Ansible

Install Ansible

-  Most Linux distributions ship two versions:
 - **ansible-core**: Ansible binary + minimal set of collections
 - **ansible**: Ansible and selected collections

Install Ansible - RHEL

-  Install Ansible on RHEL 10

```
subscription-manager repos --enable codeready-builder-for-rhel-10-$(arch)-rpms
dnf install https://dl.fedoraproject.org/pub/epel/epel-release-latest-10.noarch.rpm
dnf install ansible-core ansible-collection-ansible-posix \
    ansible-collection-community-crypto ansible-collection-community-general
```

⚙️ Install Ansible – RHEL derivatives

- ⚙️ Install Ansible on AlmaLinux or Rocky Linux 10

```
dnf config-manager --set-enabled crb
dnf install epel-release
dnf install ansible-core ansible-collection-ansible-posix \
    ansible-collection-community-crypto ansible-collection-community-general
```

Install Ansible – openSUSE

-  Install Ansible on openSUSE

```
zypper install ansible
```

Install Ansible – Ubuntu

-  Install Ansible on Ubuntu 24.04+

```
apt update  
apt install ansible
```

Install Ansible – pip

-  Install Ansible with pip

```
pip install ansible
```



LAB 2: Install Ansible

LAB 2: Install Ansible

-  Ensure Ansible is installed

```
ansible --version
```

```
ansible [core 2.16.16]
```

NOTE

Depending on your operating system the Ansible version can be different

 Prepare Ansible inventory

Prepare Ansible inventory

-  Create folder for your Ansible code

```
mkdir playbooks
cd playbooks
<EDITOR> hosts
cmk-server ansible_connection=local
cmk-agent ansible_connection=local
```

NOTE

Replace <EDITOR> with vi, nano or your favorite editor



Test connection to target server

-  Make sure your target server is reachable

```
ansible all -i hosts -m ping
cmk-server | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
cmk-agent | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
```

Ansible does not send an ICMP ping to test the connection, it tries to log into the target machine.



LAB 3: Add target to Ansible inventory

LAB 3: Add target to Ansible inventory

-  Make sure your target server is reachable

```
$ ansible all -i hosts -m ping
cmk-server | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
cmk-agent | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
```



Install Checkmk Collection

Install Checkmk Collection

-  Simplified and consistent content delivery
-  The collection contains:
 - Modules
 - Playbooks
 - Roles
 - Plugins
 - Docs
 - Tests

Install Checkmk Collection

-  Development:
<https://github.com/Checkmk/ansible-collection-checkmk.general>
-  Hosted at Galaxy:
<https://galaxy.ansible.com/ui/repo/published/checkmk/general>
-  Install the collection:

```
ansible-galaxy collection install checkmk.general
```




Installation of Checkmk server



Installation of Checkmk server

- 📦 Can be installed with the provided server role
- 💡 Strongly recommend writing your own role to fit your Ansible environment and coding guidelines!
- 📖 Role documentation:
<https://galaxy.ansible.com/ui/repo/published/checkmk/general/content/role/server/>



Installation of Checkmk server

-  Create playbook

```
<EDITOR> cmk_server.yml
```



Installation of Checkmk server

- name: Install Checkmk server
 - hosts: cmk-server
 - connection: local
 - become: true
- roles:
 - checkmk.general.server



Installation of Checkmk server

-  Create required variables

```
mkdir host_vars  
<EDITOR> host_vars/cmks-server.yml
```



Installation of Checkmk server

```
---
checkmk_server_version: "2.5.0"
checkmk_server_edition: "community"
checkmk_admin_pw: "AnsibleW0rkshop2026!"
checkmk_server_sites:
  - name: "master"
    version: "{{ checkmk_server_version }}"
    update_conflict_resolution: "abort"
    state: started
    admin_pw: "{{ checkmk_admin_pw }}"
```



LAB 4: Install Checkmk server



LAB 4: Install Checkmk server

-  Ensure Checkmk server is installed

```
ansible-playbook -i hosts cmk_server.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.16.16
```

```
PLAY [Install Checkmk server] *****
```

```
TASK [Gathering Facts] *****
ok: [cmk-server]
```

```
TASK [checkmk.general.server : Flush Handlers.] *****
```

```
...
```

```
PLAY RECAP *****
cmk-server : ok=20 changed=8 unreachable=0 failed=0 skipped=21 rescued=0 ignored=0
```




Installation of Checkmk server

-  Ensure HTTPS is configured

```
<EDITOR> cmk_server.yml
```



Installation of Checkmk server

```
...
tasks:
  - name: Install mod_ssl for https access
    ansible.builtin.dnf:
      name: mod_ssl
      state: installed
      notify: Restart httpd

handlers:
  - name: Restart httpd
    ansible.builtin.service:
      name: httpd
      state: restarted
```



LAB 5: Secure Checkmk server



LAB 5: Secure Checkmk server

-  Ensure Checkmk server is listening to HTTPS

```
ansible-playbook -i hosts cmk_server.yml
```

```
PLAY RECAP *****
cmk-server : ok=17 changed=3 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```

-  Access your server: https://<ip_address>/master
-  Username: cmkadmin
-  Password: AnsibleW0rkshop2026!





Installation of Checkmk agent



Installation of Checkmk agent

- 📦 Can be installed with the provided agent role
- 💡 Strongly recommend writing your own role to fit your Ansible environment and coding guidelines!
- 📖 Role documentation:
<https://galaxy.ansible.com/ui/repo/published/checkmk/general/content/role/agent/>



Installation of Checkmk agent

-  Create playbook

```
<EDITOR> cmk_agent.yml
```




Installation of Checkmk agent

```
---  
  
- name: Install Checkmk agent  
  hosts: cmk-agent  
  connection: local  
  become: true  
  
  pre_tasks:  
    - name: Install required Python packages  
      ansible.builtin.dnf:  
        name: python3-netaddr  
        state: installed  
  
  roles:  
    - checkmk.general.agent
```

Installation of Checkmk agent

-  Create required variables

```
mkdir host_vars  
<EDITOR> host_vars/cmka-agent.yml
```



Installation of Checkmk agent

```
---  
checkmk_agent_version: "2.5.0"  
checkmk_agent_edition: "community"  
checkmk_agent_server: localhost  
checkmk_agent_server_validate_certs: "false"  
checkmk_agent_site: "master"  
checkmk_agent_user: "cmkadmin"  
checkmk_agent_pass: "AnsibleW0rkshop2026!"
```



LAB 6: Install Checkmk agent



LAB 6: Install Checkmk agent

-  Ensure Checkmk agent is installed

```
$ ansible-playbook -i hosts cmk_agent.yml

[WARNING]: Collection checkmk.general does not support Ansible version 2.16.16

PLAY [Install Checkmk agent] *****

TASK [Gathering Facts] *****
ok: [cmk-agent]

...

TASK [checkmk.general.agent : Update monitored services and labels on host.] *****
ok: [cmk-agent]

PLAY RECAP *****
cmk-agent : ok=19 changed=3 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



Checkmk server configuration



Checkmk server configuration: Folders



Checkmk configuration: Folders

-  Create two folders - **/linux** and **/linux/checkmk**
-  Add a **label** to folder linux and ensure **Checkmk agent** is used for monitoring

```
<EDITOR> host_vars/cmkn-server.yml
```




Checkmk configuration: Folders

```
...
checkmk_folders:
- path: /linux
  attributes:
    labels:
      os: almalinux
      tag_agent: "cmk-agent"
- path: /linux/checkmk
```

NOTE

Create the folder in the Web UI first and access the API to get syntax for attributes!



Checkmk configuration: Folders

-  Create two folders - **/linux** and **/linux/checkmk**
- **+** Add a **label** to folder linux and ensure **Checkmk agent** is used for monitoring
- **+** Append a new task to the existing tasks

```
<EDITOR> cmk_server.yml
```



Checkmk configuration: Folders

```
...  
- name: Create folder  
checkmk.general.folder:  
  name: "{{ item['name'] | default(omit) }}"  
  path: "{{ item['path'] }}"  
  attributes: "{{ item['attributes'] | default(omit) }}"  
  state: present  
  server_url: "https://localhost"  
  site: "master"  
  automation_user: "cmkadmin"  
  automation_secret: "{{ checkmk_admin_pw }}"  
  validate_certs: false  
  notify: "Activate Checkmk changes"  
  loop: "{{ checkmk_folders }}"
```



Checkmk configuration: Folders

- + Append a new handler

```
...  
- name: Activate Checkmk changes  
  checkmk.general.activation:  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_admin_pw }}"  
    validate_certs: false  
    force_foreign_changes: true  
    run_once: true
```



LAB 7: Checkmk configuration: Folders

LAB 7: Checkmk configuration: Folders

-  Ensure folder is created

```
ansible-playbook -i hosts cmk_server.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.16.16
```

```
PLAY [Install Checkmk server] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [cmk-server]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-server : ok=19 changed=3 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



Checkmk server configuration: Hosts



Checkmk configuration: Hosts

-  Create host **cmk-agent** in folder **/linux/checkmk** with IP address **127.0.0.1**

```
<EDITOR> host_vars/cmk-agent.yml
```




Checkmk configuration: Hosts

```
...  
checkmk_host_settings:  
  folder: /linux/checkmk  
  attributes:  
    ipaddress: "127.0.0.1"
```

NOTE

Create the host in the Web UI first and access the API to get syntax for attributes!



Checkmk configuration: Hosts

-  Create host **cmk-agent** in folder **/linux/checkmk** with IP address **127.0.0.1**
- **+** Add a new task

```
$ <EDITOR> cmk_agent.yml
```



Checkmk configuration: Hosts

```
...
tasks:
- name: Create host
  checkmk.general.host:
    name: "{{ inventory_hostname }}"
    folder: "{{ checkmk_host_settings['folder'] | default(omit) }}"
    attributes: "{{ checkmk_host_settings['attributes'] | default(omit) }}"
    state: present
    server_url: "https://localhost"
    site: "master"
    automation_user: "cmkadmin"
    automation_secret: "{{ checkmk_agent_pass }}"
    validate_certs: false
  notify: "Activate Checkmk changes"
  loop: "{{ checkmk_host_settings | dict2items }}"
```



Checkmk configuration: Hosts

- + Append a new handler

```
...
handlers:
- name: Activate Checkmk changes
  checkmk.general.activation:
    server_url: "https://localhost"
    site: "master"
    automation_user: "cmkadmin"
    automation_secret: "{{ checkmk_agent_pass }}"
    validate_certs: false
    force_foreign_changes: true
    run_once: true
```



LAB 8: Checkmk configuration: Hosts



LAB 8: Checkmk configuration: Hosts

-  Ensure host is created

```
$ ansible-playbook -i hosts cmk_agent.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.16.16
```

```
PLAY [Install Checkmk agent] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [cmk-agent]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-agent : ok=21 changed=2 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



Checkmk server configuration: Service discovery

Checkmk configuration: Discovery

-  Run service discovery and add **unmonitored** services
- **+** Append a task to the existing tasks

```
<EDITOR> cmk_agent.yml
```


Checkmk configuration: Discovery

```
...  
- name: Run service discovery  
  checkmk.general.discovery:  
    host_name: "{{ inventory_hostname }}"  
    state: new  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_agent_pass }}"  
    validate_certs: false  
    notify: "Activate Checkmk changes"
```



LAB 9: Checkmk configuration: Service discovery

LAB 9: Checkmk configuration: Discovery

-  Ensure services are monitored

```
ansible-playbook -i hosts cmk_agent.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.16.16
```

```
PLAY [Install Checkmk agent] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [cmk-agent]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-agent : ok=21 changed=2 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```





Checkmk server configuration: Users

Checkmk configuration: Users

-  Create user **cmkuser** with email address **cmkuser@localhost** and member of contact group **Everything** and role **Normal user**

```
<EDITOR> host_vars/cmk-server.yml
```

Checkmk configuration: Users

```
...
checkmk_users:
  - name: cmkuser
    password: "AnsibleW0rkshop2026!"
    email: "cmkuser@localhost"
    fullname: "Checkmk user"
    auth_type: password
    contactgroups:
      - "all"
    roles:
      - "user"
```

Checkmk configuration: Users

-  Create user **cmkuser** with email address **cmkuser@localhost** and member of contact group **Everything** and role **Normal user**
-  Add a new task after the existing tasks

```
<EDITOR> cmk_server.yml
```


Checkmk configuration: Users

```
...
- name: Create user
  checkmk.general.user:
    name: "{{ item['name'] }}"
    password: "{{ item['password'] | default(omit) }}"
    email: "{{ item['email'] | default(omit) }}"
    fullname: "{{ item['fullname'] | default(omit) }}"
    auth_type: "{{ item['auth_type'] }}"
    contactgroups: "{{ item['contactgroups'] | default(omit) }}"
    roles: "{{ item['roles'] | default(omit) }}"
    state: present
    server_url: "https://localhost"
    site: "master"
    automation_user: "cmkadmin"
    automation_secret: "{{ checkmk_admin_pw }}"
    validate_certs: false
    notify: "Activate Checkmk changes"
    loop: "{{ checkmk_users }}"
```



LAB 10: Checkmk configuration: Users

LAB 10: Checkmk configuration: Users

-  Ensure user is created

```
$ ansible-playbook -i hosts cmk_server.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.16.16
```

```
PLAY [Install Checkmk server] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [cmk-server]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-server : ok=19 changed=3 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



Checkmk server configuration: Downtimes

Checkmk configuration: Downtimes

-  Create downtime for **15 minutes**

```
<EDITOR> cmk_downtime.yml
```

Checkmk configuration: Downtimes

```
- name: Configure Checkmk downtime
  hosts: cmk-agent
  connection: local
  become: true

  tasks:
    - name: Create downtime
      checkmk.general.downtime:
        host_name: "{{ inventory_hostname }}"
        end_after:
          minutes: 15
        state: present
        server_url: "https://localhost"
        site: "master"
        automation_user: "cmkadmin"
        automation_secret: "{{ checkmk_agent_pass }}"
        validate_certs: false
```



LAB 11: Checkmk configuration: Downtimes



LAB 11: Checkmk configuration: Downtimes

-  Ensure downtime is created

```
$ ansible-playbook -i hosts cmk_downtime.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.16.16
```

```
PLAY [Configure Checkmk downtime] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [cmk-agent]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-agent : ok=3 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
```




Checkmk server configuration: Rules

Checkmk configuration: Rules

-  Create a rule that sets the **Checkmk Agent** check to **warning** if the version is **different** from master and **critical** if TLS is **not activated**

```
<EDITOR> host_vars/cmk-server.yml
```

Checkmk configuration: Rules

```
...
checkmk_rules:
- ruleset: "checkgroup_parameters:agent_update"
  rule:
    properties:
      description: "Checkmk Agent version"
    location:
      folder: "/linux"
    value_raw: "{ 'agent_version': ('site', {}), 'agent_version_mismatch': 1, 'legacy_pull_mode': 2}"
```

NOTE

Create the rule in the Web UI first and access the API to get syntax for rule values!

Checkmk configuration: Rules

-  Create a rule that sets the **Checkmk Agent** check to **warning** if the version is **different** from master and **critical** if **TLS** is **not activated**
- **+** Add a task after the existing tasks

```
$ <EDITOR> cmk_server.yml
```



Checkmk configuration: Rules

```
...  
- name: Create rule  
  checkmk.general.rule:  
    rule: "{{ item['rule'] }}"  
    ruleset: "{{ item['ruleset'] }}"  
    state: present  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_admin_pw }}"  
    validate_certs: false  
    notify: "Activate Checkmk changes"  
    loop: "{{ checkmk_rules }}"
```



LAB 12: Checkmk configuration: Rules

LAB 12: Checkmk configuration: Rules

-  Ensure the rule for agent monitoring is created

```
ansible-playbook -i hosts cmk_server.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.16.16
```

```
PLAY [Install Checkmk server] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [cmk-server]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-server : ok=20 changed=3 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



Checkmk server configuration: Passwords

Checkmk configuration: Passwords

-  Create a password for the site **authenticationtest.com**

```
<EDITOR> host_vars/cmkn-server.yml
```

Checkmk configuration: Passwords

```
...
```

```
checkmk_passwords:  
- name: authenticationtest_com  
  title: authenticationtest.com  
  password: pass
```

Checkmk configuration: Passwords

-  Create a password for the site **authenticationtest.com**
- **+** Add tasks **before the Create rule** task!

```
<EDITOR> host_vars/cmkn-server.yml
```



Checkmk configuration: Passwords

...

```
- name: Create password
checkmk.general.password:
  name: "{{ item['name'] }}"
  title: "{{ item['title'] | default(omit) }}"
  password: "{{ item['password'] }}"
  state: present
  server_url: "https://localhost"
  site: "master"
  automation_user: "cmkadmin"
  automation_secret: "{{ checkmk_admin_pw }}"
  validate_certs: false
notify: "Activate Checkmk changes"
loop: "{{ checkmk_passwords }}"
```

Checkmk configuration: Passwords

-  Create a password for the site **authenticationtest.com**
-  Add the rule to the existing rules

```
<EDITOR> host_vars/cmkn-server.yml
```



Checkmk configuration: Passwords

```
...
checkmk_rules:
...
- ruleset: "active_checks:httpv2"
  rule:
    properties:
      description: "authenticationtest.com"
    location:
      folder: "/"
    value_raw: >
      "{ 'endpoints': [{ 'service_name': { 'prefix': 'auto', 'name': 'HTTPAuth'}, 'url': 'https://authenticationtest.com/HTTPAuth/' }],
        'standard_settings': { 'connection': { 'method': ('get', None), 'auth': ('user_auth', { 'user': 'user', 'password':
          ('cmk_postprocessed', 'stored_password', ('authenticationtest_com', ''))) } } } } }"
    conditions:
      host_name:
        match_on:
          - cmk-agent
        operator: one_of
```

NOTE

Create the rule in the Web UI first and access the API to get syntax for rule values!



LAB 13: Checkmk configuration: Passwords

LAB 13: Checkmk configuration: Passwords

-  Ensure the rule and password for web monitoring are created

```
$ ansible-playbook -i hosts cmk_server.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.16.16
```

```
PLAY [Install Checkmk server] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [cmk-server]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-server : ok=21 changed=4 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```






Fetch information from Checkmk with lookup plugins

Checkmk lookup

-  Print **server version** of Checkmk server
- **+** Add a task after the existing tasks

```
<EDITOR> cmk_server.yml
```

Checkmk lookups

...

- name: Print Checkmk server version

ansible.builtin.debug:

```
msg: "Checkmk server version is {{ lookup('checkmk.general.version',  
server_url='https://localhost',  
site='master',  
automation_user='cmkadmin',  
automation_secret=checkmk_admin_pw,  
validate_certs=false)}}
```

```
}}"
```



LAB 14: Checkmk lookups

LAB 14: Checkmk lookups

-  Print Checkmk server version

```
$ ansible-playbook -i hosts cmk_server.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.16.16
```

```
PLAY [Install Checkmk server] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [cmk-server]
```

```
...
```

```
TASK [Print Checkmk server version] *****
```

```
ok: [cmk-server] => {
```

```
  "msg": "Checkmk server version is 2.5.0.cre"
```

```
}
```

```
PLAY RECAP *****
```

```
cmk-server : ok=22 changed=3 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



Using Checkmk as Ansible inventory



Checkmk as Ansible inventory

-  Checkmk **should** be aware of all hosts
-  Can be a good source for Ansible
-  Most likely requires tag configuration



Checkmk as Ansible inventory

-  Create tags for **testing** and **production** hosts

```
<EDITOR> host_vars/cmkn-server.yml
```

Checkmk as Ansible inventory

```
...
```

```
checkmk_tag_groups:  
  - name: lifecycle_environment  
    title: Lifecycle Environment  
    tags:  
      - id: production  
        title: Production  
      - id: testing  
        title: Testing
```

Checkmk as Ansible inventory

-  Create tags for **testing** and **production** hosts
- **+** Add a task after the existing tasks

```
<EDITOR> host_vars/cmkn-server.yml
```



Checkmk as Ansible inventory

```
...  
- name: Create tag group  
  checkmk.general.tag_group:  
    name: "{{ item['name'] }}"  
    title: "{{ item['title'] }}"  
    tags: "{{ item['tags'] }}"  
    state: present  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_admin_pw }}"  
    validate_certs: false  
    notify: "Activate Checkmk changes"  
    loop: "{{ checkmk_tag_groups }}"
```

Checkmk as Ansible inventory

-  Create an inventory config for Checkmk

```
<EDITOR> checkmk.yml
```

NOTE

Inventory plugin must end with checkmk.yml or checkmk.yaml!



Checkmk as Ansible inventory

```
plugin: checkmk.general.checkmk
server_url: https://localhost
site: master
automation_user: cmkadmin
automation_secret: "AnsibleW0rkshop2026!"
validate_certs: false
groupsources:
  - hosttags
  - sites
```



LAB 15: Checkmk as Ansible inventory

LAB 15: Checkmk as Ansible inventory

-  Ensure tag groups are created

```
ansible-playbook -i hosts cmk_server.yml
```

```
[WARNING]: Collection checkmk.general does not support Ansible version 2.16.16
```

```
PLAY [Install Checkmk server] *****
```

```
TASK [Gathering Facts] *****
```

```
ok: [cmk-server]
```

```
...
```

```
PLAY RECAP *****
```

```
cmk-server : ok=23 changed=4 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```




LAB 15: Checkmk as Ansible inventory

-  Print all hosts and groups

```
ansible-inventory -i checkmk.yml --graph
```

```
@all:
  |--@ungrouped:
  |--@tag_criticality_prod:
  |   |--cmk-agent
  ...
  |--@tag_lifecycle_environment_production:
  |   |--cmk-agent
  |--@tag_lifecycle_environment_testing:
  ...
  |--@tag_lifecycle_environment_production:
  |   |--cmk-agent
  |--@tag_lifecycle_environment_testing:
```



LAB 15: Checkmk as Ansible inventory

-  Print host details

```
ansible-inventory -i checkmk.yml --host cmk-agent
```

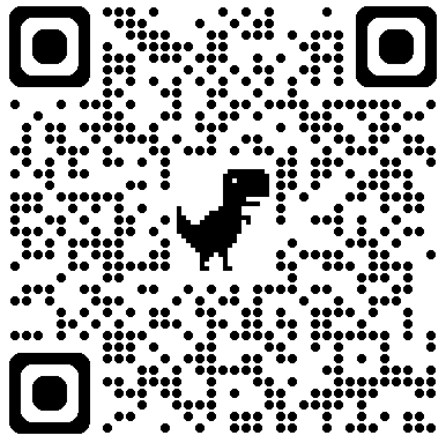
```
{
  "checkmk_agent_edition": "cre",
  "checkmk_agent_pass": "AnsibleW0rkshop2026!",
  "checkmk_agent_server": "localhost",
  "checkmk_agent_server_validate_certs": "false",
  "checkmk_agent_site": "master",
  "checkmk_agent_user": "cmkadmin",
  "checkmk_agent_version": "2.5.0",
  "checkmk_host_settings": {
    "attributes": {
      "ipaddress": "127.0.0.1"
    },
    "folder": "/linux/checkmk"
  },
  "folder": "/linux/checkmk",
  "ipaddress": "127.0.0.1"
}
```


 **Demo: Self-healing with Checkmk, Ansible
Automation Platform and rulebooks**

 Feedback

Feedback

- <https://forms.gle/ki6KjQF9KS6BavmU8>





Thank you for participating

- DI (FH) René Koch
- Freelancer
- The Checkmk Conference #12, 18.06.2026